Lateral OFC represents reward identity during conditioning

Results

The preferential representation of preconditioned outcomes in lateral OFC may reflect the prioritization of inferred outcome identities when both cue classes share limited representational resources. We therefore asked whether directly conditioned cues would evoke odor-related representations during conditioning, when only conditioned cues were presented and participants explicitly reported which odor would follow each cue (Fig. XA). Using the same independently defined lateral and medial OFC regions, we tested whether cues predicting the same food odor evoked more similar neural patterns than cues predicting different odors. RSA used single-trial activity estimates spanning the combined cue and prediction period (0–5 s), pooled correct trials across all available conditioning runs ($n = 29$), and excluded clean-air trials. Pattern comparisons were restricted to presentations from different runs. Neural patterns in lateral OFC showed a small positive correspondence with the odor-identity model (mean Spearman $\rho = 0.0131$, $t(28) = 2.01$, $p = 0.027$), whereas the corresponding effect was not significant in medial OFC (mean Spearman $\rho = 0.0068$, $t(28) = 1.20$, $p = 0.120$; Fig. XB).

To test whether conditioning cues also evoked patterns shared with experienced odors, we trained the same cross-task decoder on responses to the two food odors presented during the independent localizer task and applied it to conditioning cue patterns from an early cue window (0–2 s; EARLY) and the combined cue and prediction window (0–5 s; EXT5s). In lateral OFC, decoding accuracy exceeded 50% chance for EARLY (52.12%, $t(28) = 1.83$, $p = 0.039$; Fig. XC). Accuracy was also numerically above chance for EXT5s, although this effect did not reach significance (51.87%, $t(28) = 1.45$, $p = 0.079$). Decoding was not significantly above chance in medial OFC for either window (EARLY, 49.25%, $p = 0.715$; EXT5s, 50.05%, $p = 0.484$).

Together, these analyses provide suggestive evidence of odor-related information in lateral OFC during explicit prediction of directly associated outcomes, consistent with previous findings of identity-specific reward representations in this region. This pattern is also consistent with the model's proposal that lateral OFC allocates outcome-identity representations according to task demands: when only conditioned cues are presented, these representations can support explicit prediction of their associated odors.

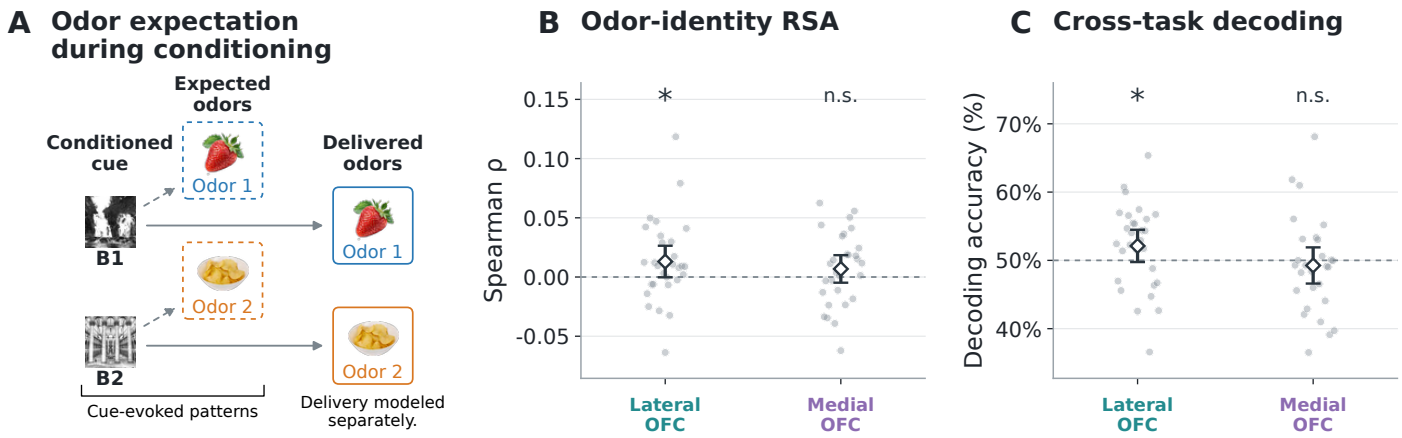


Figure X. Lateral OFC represents reward identity during conditioning. A. Odor expectation during conditioning. Dashed branches link conditioned cues to their expected odors, shown in dashed boxes; solid boxes indicate delivered odors. Odor images are illustrative and were not displayed during the task. Odor delivery was modeled separately in the single-trial GLMs. **B. Odor-identity RSA.** Spearman correlations between neural and expected odor-identity RDMs were computed from cue and prediction-period estimates (0–5 s; EXT5s). **C. Cross-task decoding.** A classifier trained on passive odor responses was tested on early conditioning cue responses (0–2 s; EARLY). In B and C, analyses used the same bilateral OFC regions as Figure 3. Points represent individual participants (n = 29), and open diamonds indicate means with t-based 95% confidence intervals. Dashed lines indicate zero RSA correlation (B) and 50% chance accuracy (C). Asterisks denote uncorrected $p < 0.05$ from one-sided one-sample t-tests against these baselines; n.s., not significant.

Methods

We estimated single-trial responses during conditioning using the same preprocessing procedures and nuisance regressors as in the probe analyses. Each least-squares-separate model contained a target-cue regressor and a second regressor pooling all other cue presentations, spanning either the first 2 s (EARLY) or the combined cue and prediction period (5 s; EXT5s). EARLY models additionally included a pooled prediction regressor (2–5 s after cue onset). Both models included delay (5–7 s), odor delivery (7–9 s), zero-duration button presses at the recorded response times, and fixation replacing delay and delivery on missed-response trials (4 s). Delivery events for both food odors and clean air were pooled into one regressor and included on all responded trials, including incorrect predictions. All cue presentations were modeled; correct-trial filtering was applied subsequently.

RSA and cross-task decoding used the same bilateral OFC regions as the probe analyses. Only correct food-odor trials with recorded responses and finite positive response times were retained, pooling all available conditioning runs. Two additional participants were excluded from the conditioning analyses. One showed poor conditioning performance across runs 1–2 (8 of 24 food-odor trials correct in run 1 and 4 of 24 in run 2), despite improved performance in run 3. The other had virtually no usable OFC voxels after excluding voxels with nonfinite estimates across runs 1–3 (0/4,445 lateral and 4/5,500 medial OFC voxels retained). This left 29 participants; 28 contributed three runs and one contributed runs 1–2 because the third scan was interrupted. RSA used EXT5s estimates and the probe-analysis procedures, computing Spearman correlations between neural and expected odor-identity RDMs using only cross-run pairs; participant-level coefficients were assessed against zero. Cross-task decoding followed the probe-analysis procedure, training a classifier on passive odor-localizer patterns and applying it to both conditioning windows without further fitting. EARLY and EXT5s used matched trials and a common set of voxels finite across training and testing; voxel responses were z-scored separately within each task, window, and run. Decoding accuracy averaged recall across the two food odors and was assessed against 50% chance. Both analyses used one-sided one-sample t-tests with uncorrected p values.